

UST'-KULOM, Russian Federation

WMO: 238030

Lat: 61.6994N

Lon: 53.6950E

Elev: 143

StdP: 99.61

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.1	-31.2	-38.0	0.1	-35.1	-34.0	0.2	-31.1	8.2	-6.9	7.2	-7.1	1.0	90	0.640

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	28.1	20.0	26.1	18.7	24.1	17.6	21.1	26.3	19.8	24.6	18.5	22.8	2.3	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.2	14.2	23.7	18.0	13.1	22.7	16.7	12.1	21.2	61.6	26.5	57.3	24.6	53.1	22.8	27.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	7.7	6.5	5.5	DB	-36.1	30.8	4.5	1.8	-39.4	32.1	-42.0	33.1	-44.6	34.1	-47.8	35.4
(5)				WB	-36.9	22.6	3.7	1.8	-39.5	23.9	-41.7	25.0	-43.8	26.0	-46.5	27.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	0.8	-15.4	-14.0	-6.4	1.2	8.1	13.5	17.0	13.3	8.1	1.3	-6.3	-11.9	(6)
(7)		DBStd	12.48	8.47	7.87	5.52	5.02	5.39	4.65	4.39	4.15	3.82	4.17	6.66	9.27	(7)
(8)		HDD10.0	3893	788	673	509	268	103	17	4	14	79	271	488	679	(8)
(9)		HDD18.3	6462	1046	906	768	516	318	157	78	164	306	529	738	938	(9)
(10)		CDD10.0	532	0	0	0	2	46	123	221	116	24	1	0	0	(10)
(11)		CDD18.3	58	0	0	0	0	2	12	36	8	0	0	0	0	(11)
(12)		CDH23.3	562	0	0	0	1	32	112	346	72	1	0	0	0	(12)
(13)		CDH26.7	119	0	0	0	0	4	21	79	15	0	0	0	0	(13)
(14)	Wind	WSAvg	2.5	2.5	2.5	2.8	2.8	2.7	2.4	2.0	2.1	2.2	2.7	2.7	2.6	(14)
(15)	Precipitation	PrecAvg	619	43	32	33	34	52	72	61	69	59	61	57	44	(15)
(16)		PrecMax	780	75	75	55	74	116	156	116	145	114	101	112	78	(16)
(17)		PrecMin	495	16	6	7	1	23	14	26	22	15	24	12	19	(17)
(18)		PrecStd	72	14	17	13	19	26	36	23	32	26	22	24	14	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.5	1.3	6.8	19.1	26.6	29.2	30.6	28.7	20.7	14.8	7.2	2.3	(19)
(20)			MCWB	0.0	0.2	2.7	10.8	15.8	20.4	21.3	21.0	15.8	11.2	6.5	1.7	(20)
(21)		2%	DB	-0.7	0.0	3.9	13.5	23.4	26.6	28.9	25.2	17.8	10.9	3.8	1.0	(21)
(22)			MCWB	-1.2	-0.6	0.8	7.4	14.3	19.2	20.8	18.4	13.7	9.1	3.2	0.4	(22)
(23)		5%	DB	-2.2	-1.6	2.2	10.6	20.5	24.2	27.1	22.7	15.9	8.7	2.0	0.0	(23)
(24)			MCWB	-2.5	-2.2	0.2	5.9	13.1	17.7	19.5	17.2	12.7	7.5	1.5	-0.4	(24)
(25)		10%	DB	-4.4	-3.6	0.8	8.3	17.7	22.1	25.2	20.3	14.0	7.2	0.8	-1.3	(25)
(26)			MCWB	-4.7	-4.2	-0.6	4.2	11.3	16.1	18.5	16.1	11.7	6.2	0.3	-1.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.1	0.5	3.3	11.7	17.4	21.9	23.3	22.1	16.3	11.9	6.4	1.8	(27)
(28)			MCDWB	0.3	0.9	5.9	18.5	24.8	27.2	28.7	27.5	19.3	13.4	7.3	2.0	(28)
(29)		2%	WB	-1.2	-0.8	1.5	8.4	15.2	20.1	21.6	19.5	14.4	9.2	3.2	0.6	(29)
(30)			MCDWB	-0.8	-0.2	3.0	12.3	21.4	25.3	27.0	23.9	16.8	10.5	3.7	0.9	(30)
(31)		5%	WB	-2.6	-2.3	0.5	6.3	13.7	18.6	20.5	17.9	13.1	7.7	1.5	-0.4	(31)
(32)			MCDWB	-2.3	-1.6	1.9	9.8	19.6	23.4	25.6	21.4	15.1	8.7	1.9	0.0	(32)
(33)		10%	WB	-4.7	-4.1	-0.5	4.7	12.1	17.1	19.3	16.6	11.9	6.2	0.4	-1.7	(33)
(34)			MCDWB	-4.4	-3.6	0.8	7.7	16.8	20.9	23.7	19.7	13.9	7.1	0.7	-1.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.3	7.4	8.4	8.7	10.1	10.0	10.5	8.8	7.3	4.4	4.5	5.7	(35)
(36)			MCDBR	6.2	5.7	8.1	12.9	14.6	12.7	13.6	12.3	10.9	6.9	4.1	3.8	(36)
(37)			MCWBR	6.2	5.2	6.1	8.3	8.0	6.9	6.5	6.6	7.0	5.4	4.0	3.8	(37)
(38)		5% WB	MCDBR	6.2	5.3	7.0	11.4	13.3	11.6	12.3	11.1	9.4	6.6	4.1	3.8	(38)
(39)			MCWBR	6.2	5.1	5.6	8.0	8.3	7.0	6.7	6.6	6.6	5.4	4.1	3.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.258	0.300	0.339	0.367	0.364	0.366	0.404	0.385	0.345	0.319	0.267	0.236	(40)	
(41)		taud	1.962	2.032	2.018	2.120	2.396	2.440	2.332	2.368	2.448	2.335	2.059	1.921	(41)	
(42)		Ebn at Noon	398	632	744	811	857	861	810	795	765	635	391	242	(42)	
(43)		Edh at Noon	42	78	113	125	104	101	110	98	75	58	37	25	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.28	0.99	2.24	3.64	4.83	5.09	5.24	3.60	2.05	0.80	0.27	0.13	(44)	
(45)		RadStd	0.04	0.21	0.33	0.49	0.48	0.39	0.45	0.35	0.27	0.09	0.06	0.02	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page